Gossip in the Microbiome

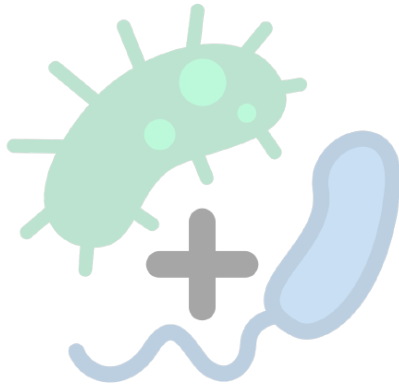
Inferring microbial interactions and how they evolve

Haris Zafeiropoulos

haris.zafr@gmail.com.
https://hariszaf.github.io

Balance of the Microverse
Friedrich Schiller University Jena

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Education and work experience

Education & training

- BSc — Biology, National & Kapodestrian University of Athens, 2016
- MSc — Bioinformatics, University of Crete, 2018
- PhD — Developing approaches for the analaysis of HTS, data integration & metabolic networks, University of Crete, 2022

Post-doctoral studies

- Lab of Microbial Systems Biology, KUL, 2022-2026
- m3i lab, University of Jena, 2026 -

In one line

Keen on **microbial interactions**, and building a **toolkit** to study those and their evolution.

Developing solutions for the analysis of HTS data

Amplicon and Metagenomics Analysis

Support large scale (meta)analyses and metadata.

PEMA

An end-to-end pipeline for **environmental DNA metabarcoding**, from raw reads to taxonomy-annotated ASVs/OTUs (12S, 16S, 18S, ITS, COI).

DARN

A **Dark mAtteR iNvestigator** for COI amplicons: places reads on a reference tree to reveal how much of the signal is non-target or unknown taxa.

METAGOFLOW

A **metagenomics workflow** (CWL) for marine Genomic Observatories, giving reproducible taxonomic and functional profiles packaged as RO-Crates.

PREGO

Extracting *Processes* - *Environments* - *Organisms* asssociations, using text mining and data integrations on public literature and global repositories (MGnify, JGI/IMG etc.)

Genome-scale metabolic models

Why metabolic models?

- Link genomes to **phenotypes**: what a microbe can take up, produce and exchange
- Constraint-based methods (FBA, flux sampling) on genome-scale reconstructions

DNNGIOR

Deep-learning-guided **gap-filling** of draft metabolic reconstructions.

Now integrated with three GEM reconstruction (ModelSEEDPy, carve, gapseq) and MEMOTE for semi-automated reconstructions.

Inferring Interactions with FBA

Prediction of growth rates with FBA using semi-curated GEMs is currently not sufficiently accurate to predict interaction strengths reliably.

dingo

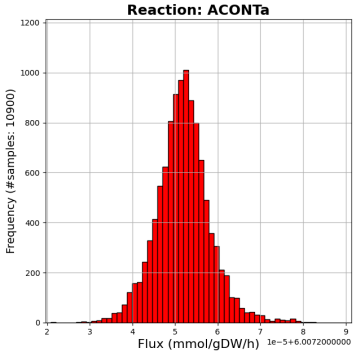
Python package for **flux sampling** and analysis of metabolic networks.

Able to sample the largest models available, with a variety of MCMC algorithms, and statistical guarantees.

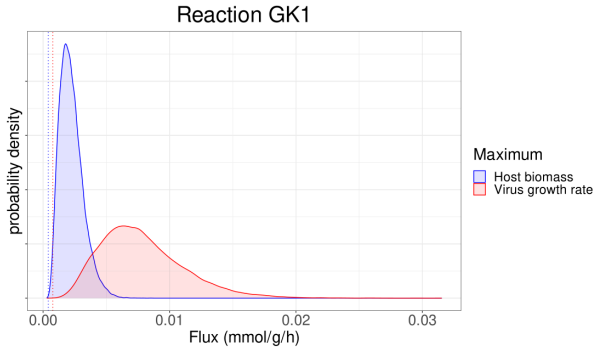
Under development: sampling community models

dingo: beyond *a* single optimum

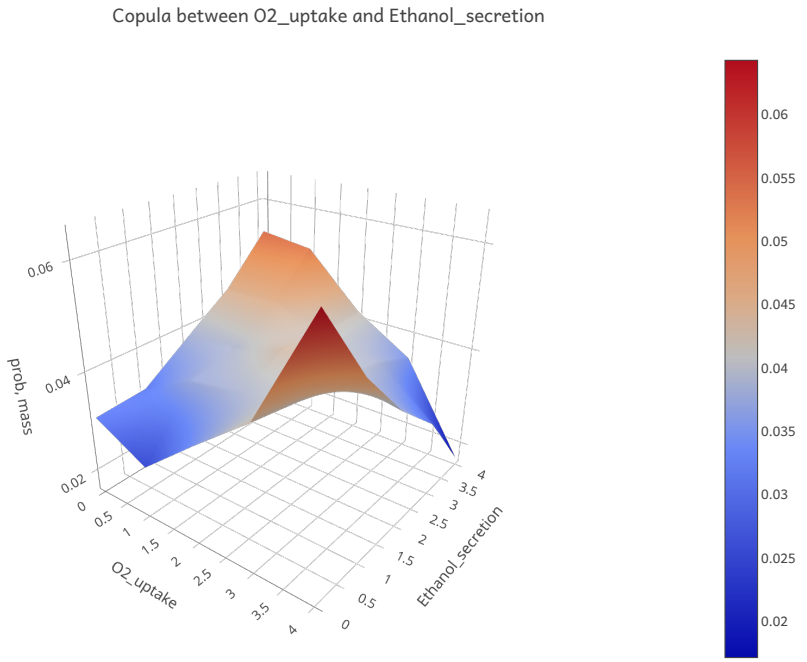
FBA gives **one** optimal flux vector vs. flux sampling explores the **whole** feasible space



Marginal: the full **distribution** of *a* reaction's flux, also in **sub-optimal** growth



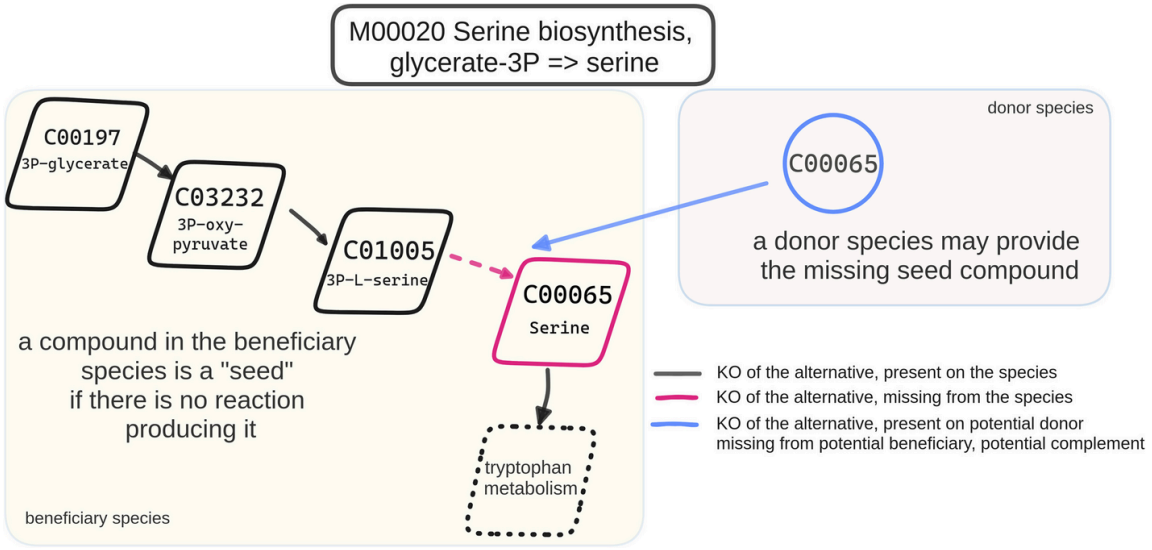
GK1: flux increases when the SARS-CoV-2 growth rate is maximised – *a* potential ***antiviral target***



Copula: O_2 uptake and ethanol secretion are **negatively** coupled – more respiration, less fermentation

microbetag

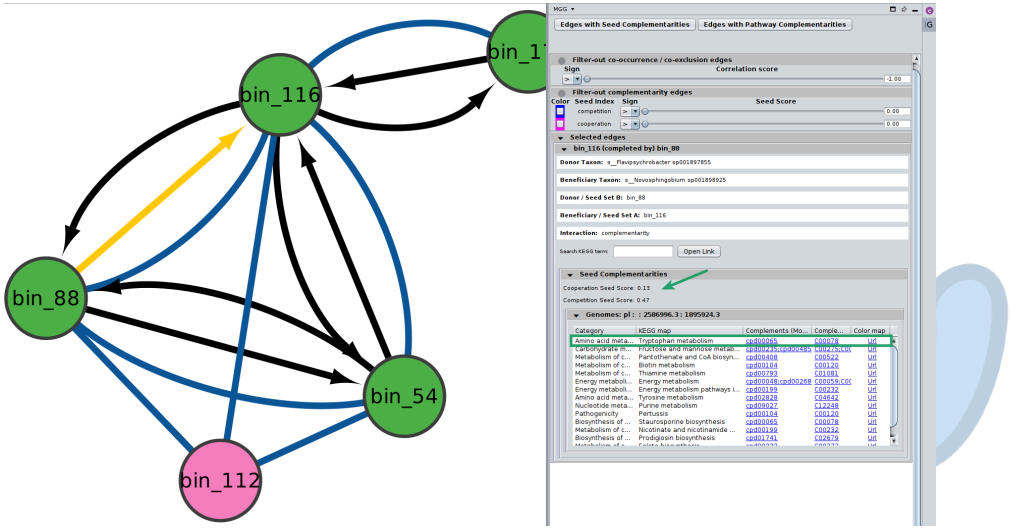
Network → **traits** **pathway complementarity** **seed scores & complementarity** → Cytoscape / MGG



Seed complementarity: *a* species' missing seed *can* be produced by *a* partner

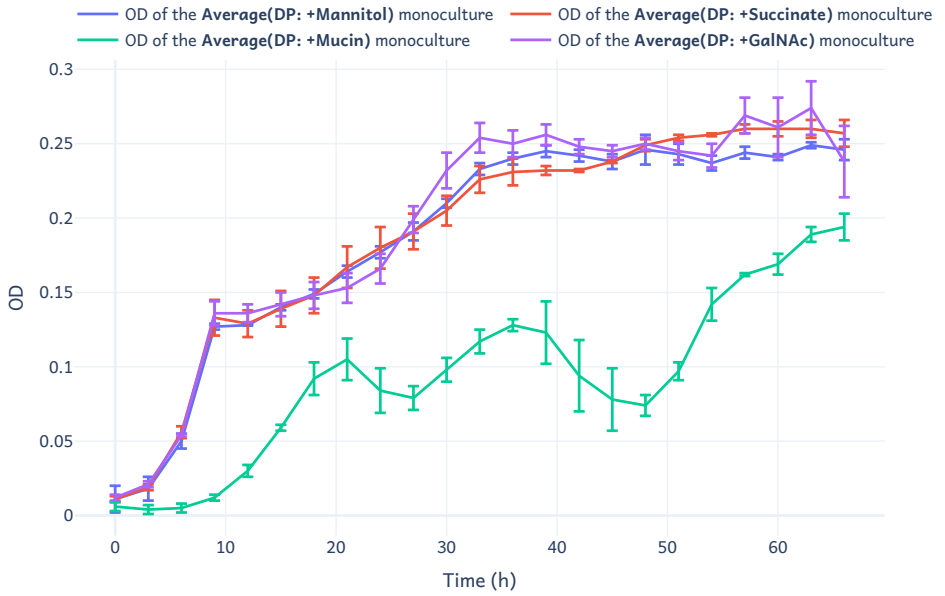
Precomputed for **GTDB v207**, or run locally (Nextflow)

Genome Biology 26:292 (2025)



mgrowthdb

- A database of **microbial growth** data (curves, conditions, metabolites)
- Makes experimental data reusable for model building and validation
- Bridges static metabolic models and **dynamics** over time



Applications

Here, you can find a list of applications that use data imported from mGrowthDB. These are third-party projects, listed as-is without endorsement. If you know of a project that integrates with mGrowthDB that is not shown here, consider reaching out to us to let us know.

mGrowthCtrl

A Python package for parameter inference, mechanistic microbiome modeling, and control, based on a consumer–resource model (CRM).

grownet

Derives interaction networks from mono-and co-culture data in mGrowthDB. The networks can be send to Cytoscape or used for gLV simulations.

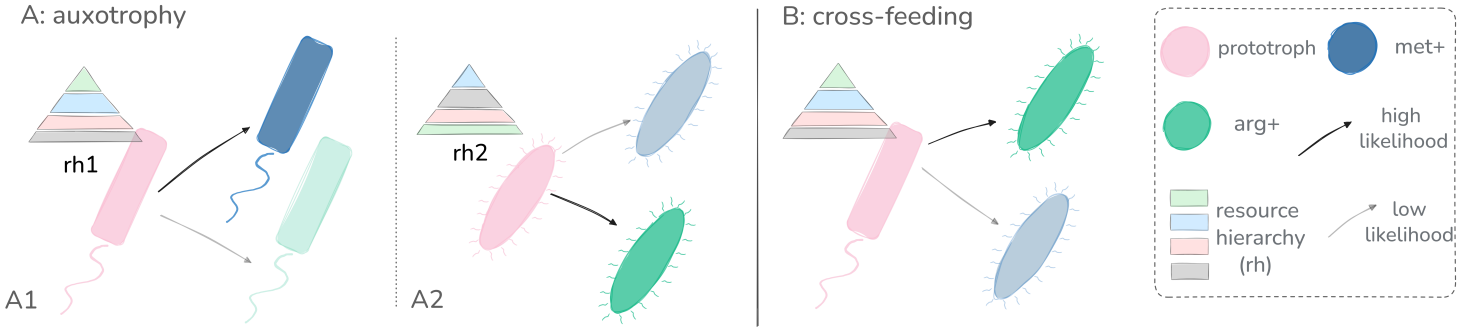
foodnet

A GUI and command line to construct metabolite consumption and production matrices for query organisms and conditions from monoculture data in mGrowthDB. The results can be downloaded as matrices or as a bipartite network and the latter can be visualized in Cytoscape.

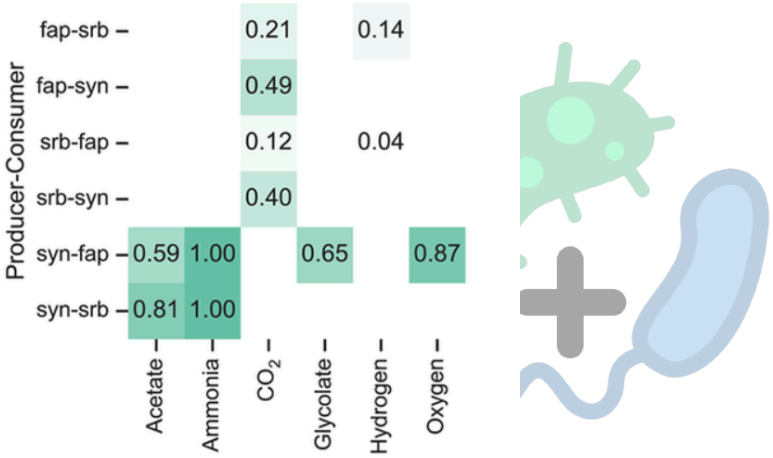
Research Interests

Theoretical Framework for Microbial Interactions

The architecture of metabolic networks as a proxy for the evolution of microbial features and their interactions



Combining topological approaches (like Minimal Pathways) with constraint-based modeling approaches (sampling) to study metabolic interactions space and probabilities and their corresponding genomes constraints



Why I'd like to join

The missing piece in my research

Metabolic networks tell me which interactions are **possible**; to test how they **evolve**, I need to know which ones **persist over time**.

What I would learn

- **Dynamic models** of communities (GLV, consumer–resource)
- **Inferring interactions** from time series (e.g. PINNs)
- **Stability and coexistence:** what selection can act on

What I would bring

- GEMs **reconstruction** (DNNGIOR) and understanding
- Flux **sampling** (dingo) GEMs, subspaces and community models
- **Interaction** priors from metabolism (microbetag)
- **Growth-curve** data for calibration (mgrowthdb)

Take-home

Metabolism says **who can** interact, dynamics say **who does**: together, they let me ask **how** interactions **evolve**.

Thanks for Listening.

I am happy to take your questions!

Slides available on GitHub

